

KnowServe Corp. — Research & Development Department

Prototype Evaluation Report — March 2025

Prototype ID: AURA-Beta-1.3

Objective

Assess system stability and retrieval accuracy for the new semantic search module.

Testing Metrics

- Average Latency – 187 ms (Goal < 200 ms) 
- Precision@10 – 0.91 (Goal ≥ 0.90)
- Recall@10 – 0.88
- Memory Footprint – 420 MB (–12 % vs previous version)
- Failure Rate – 0.3 % of queries.

Evaluation Method

10,000 synthetic queries run on a 500k-document corpus using AWS t3.medium instances.

Load testing executed via Locust; results aggregated in Grafana dashboard `/rd/aura-metrics`.

Findings

Stability acceptable for internal rollout. Minor memory leaks under high concurrency (> 1,000 req/s).

Recommend optimizing index serialization and adding batch cache layer.

Next Steps

- Fix memory leak by April 10.
- Conduct security review (Q2).
- Plan external pilot for Enterprise clients (Q3).